

Experimental Studies of Evolution and Eco-Evo Dynamics in Guppies (*Poecilia reticulata*)

David N. Reznick¹ and Joseph Travis²

¹Department of Evolution, Ecology and Organismal Biology, University of California, Riverside, California 92521, USA; email: david.reznick@ucr.edu

²Department of Biological Sciences, Florida State University, Tallahassee, Florida 32306, USA; email: travis@bio.fsu.edu

Annu. Rev. Ecol. Evol. Syst. 2019. 50:335–54

First published as a Review in Advance on
August 15, 2019

The *Annual Review of Ecology, Evolution, and
Systematics* is online at ecolsys.annualreviews.org

<https://doi.org/10.1146/annurev-ecolsys-110218-024926>

Copyright © 2019 by Annual Reviews.
All rights reserved

Keywords

Poecilia reticulata, *P. reticulata*, experimental evolution, life history evolution, eco-evo dynamics, density-dependent selection, contemporary evolution

Abstract

Guppies in Trinidad range across aquatic environments with fish communities that vary in risk of predation. These communities are often discrete, separated by waterfalls, with high-predation communities downstream and low-predation communities upstream. This gradient is repeated in many rivers; in each one, we see the same divergence between guppy populations in life history, behavior, morphology, and physiology. We have shown that the agent of selection on the life history, behavior, and physiology in low-predation communities is high population density and the cascade of ecological effects that stems from it. In effect, guppy populations modify their ecosystem and, in so doing, impose selection on themselves and shape their own evolution, which further changes the ecosystem. Evolution unfolds rapidly in this system, which has enabled us to study the dynamics of the process, not just its end points. Those studies enable us to answer some very general questions in ecology and evolutionary biology.

**ANNUAL
REVIEWS CONNECT**

www.annualreviews.org

- Download figures
- Navigate cited references
- Keyword search
- Explore related articles
- Share via email or social media

1. INTRODUCTION

How, why, and how fast does adaptive evolution happen in the real world? Our work on the Trinidadian guppy is built around addressing these questions and, more precisely, testing predictions of evolutionary theory with experiments performed in nature. We chose the path of experimental studies of evolution in nature because, at the time (1978), it was a path rarely traveled and our goal was to see how evolution works in the complexity of the real world, rather than in a laboratory setting. We built the project around testing predictions from the theory of life history evolution because the topic was compelling and the theory provided clear, testable predictions. We chose guppies because they displayed the population variation in mortality risk necessary to test theory and they occurred in a setting that made those tests feasible.

The early success of our experiments produced, as a by-product, the discovery that evolution in nature could be far more rapid, by several orders of magnitude, than had been inferred from the fossil record. Others were making the same discovery (Hendry & Kinnison 1999), which led to profound changes in multiple disciplines in how evolution was incorporated into theory and practice (Reznick et al. 2019b). In our own research, this meant thinking of ecology and evolution as contemporary, interacting processes, which makes their joint dynamics capable of producing different results than one would predict under a more traditional, historical view of evolution.

Our ongoing research is a continuing exploration of both the drivers and the consequences of contemporary evolution. We have expanded beyond the initial focus on life histories to consider how evolution shapes single-species population regulation, multi-species communities, and ecosystem processes. We have also become interested in other fundamental questions: How fast and how repeatable is evolution? Does the speed of evolution inform us about the underlying genetics, specifically the role of standing genetic variation? Are the adaptations we see in natural populations driven by a single, dominant agent of selection, or multiple, potentially interacting agents? How much ecology do you need to understand adaptive evolution? Conversely, how much evolution is necessary to understand ecology? Finally, can we understand evolution simply by studying the footprints of adaptation in living organisms, or is there something to be gained by also considering the dynamics of the process? In this review, we show how our work is answering these questions.

Our research program is built on the pioneering work of Caryl Haskins (Haskins et al. 1961), who originally described the diverse fish communities where guppies occur in the Northern Range Mountains of Trinidad. He documented how those communities were distributed throughout the many streams that drain the seasonal rain forest covering this mountain range, and he characterized guppy predators. This article was the inspiration for all who came after him.

2. BACKGROUND

2.1. Guppies

Guppies (*Poecilia reticulata*) are small (adult sizes 15–25 mm) live-bearing fish in the family Poeciliidae. Their native range includes freshwater streams in northeast South America and associated continental islands. Males are smaller than females, display a diversity of bright color patterns, and engage in elaborate courtship. Females give birth to precocial young that receive no postnatal care. Ages at maturity in the laboratory are two to three months. Females reproduce continuously thereafter and produce a brood of offspring approximately every 25 days. In nature, reproduction slows during the wet season, when food is scarce because floods scour the streams of the benthic food sources on which guppies rely. Generation times in nature are approximately 100 days in complex fish communities and 180–200 days in the smaller, headwater streams (Reznick et al. 1997).

2.2. Fish Communities

The Northern Range Mountains offer a natural laboratory for studying evolution in action. The rivers draining these mountains flow over steep gradients punctuated by waterfalls that define boundaries between different fish communities. Species diversity decreases upstream because waterfalls block the upstream dispersal of larger, predaceous fish species. The succession from diverse communities downstream to simpler communities upstream is repeated in multiple rivers, providing natural replication. In addition, fish can be introduced into portions of streams bracketed by waterfalls, creating in situ experiments (Endler 1978, 1980).

Guppies are found throughout most of these communities. In the downstream localities, they co-occur with a diversity of predators, which prey on adult guppies (high predation). Waterfalls often exclude predators but not guppies, so guppies found above waterfalls have greatly reduced risks of predation and increased life expectancy (low predation). *Rivulus bartii*, the only other fish found in these localities, rarely preys on guppies. When it does, it preys predominantly on small, immature size classes (Endler 1978, Haskins et al. 1961). In some headwater streams, *R. bartii* is the only fish species present because individuals are capable of overland travel on rainy nights.

A second form of replication occurs among these rivers. During past glaciations, when sea levels were lower, Trinidad was joined to South America, and some rivers on the south slope were tributaries to others in South America. The predators on the south slope are cichlids and characins, the same as those in South American freshwater streams. The rivers of the north slope never had mainland connections. The downstream predators are instead mostly gobies that invaded from the sea. Nevertheless, the contrast between high and low predation, downstream to upstream, is repeated on both sides of the Northern Range.

The spatial scale of this variation is small. All of our sites in the Northern Range fit within a rectangle that is approximately 25 km from west to east and 15 km from south to north. The elevational range of guppies runs from sea level to a few hundred meters. The distance between high- and low-predation communities within a river is generally in the range of hundreds of meters but can also be defined by a single barrier waterfall, with a high-predation community below the fall and a low-predation community above it. The small scale of variation means that differences in fish communities are not confounded with differences in the physical environment. The rivers are small, as well. Most of our study sites are small enough to step or jump over, and many are subdivided into discrete pools and riffles, which enables us to follow every fish in the population. Figure 1a,b illustrates the distribution of streams, and their high- and low-predation communities.

2.3. Molecular Genetics and Phylogeography

Investigators have detailed the patterns of genetic variation among and within guppy populations throughout the Northern Range Mountains using allozymes (Carvalho et al. 1991, Shaw et al. 1991), microsatellites (Barson et al. 2009, Crispo et al. 2006, Suk & Neff 2009), nucleotide base pair sequencing (Alexander et al. 2006, Fajen & Breden 1992), and genome-wide scans of single nucleotide polymorphisms (SNPs) (Fraser et al. 2015, Willing et al. 2010). All of these methods point to the same conclusions, outlined here.

The Northern Range Mountains can be divided into three biogeographic regions. The southern slopes are drained by rivers that are tributaries to either the Caroni River, which flows to the west, or the Oropouche River, which flows to the east. These rivers share fish communities dominated by cichlids and characins. West (Caroni River) guppies are most closely related to those from the Orinoco Basin of Venezuela (Alexander et al. 2006). Those from east Trinidad are widely divergent from those in west Trinidad, so much so that they are regarded by some as a different species—*Poecilia obscura* (Schories et al. 2009). The data available thus far do not link the east

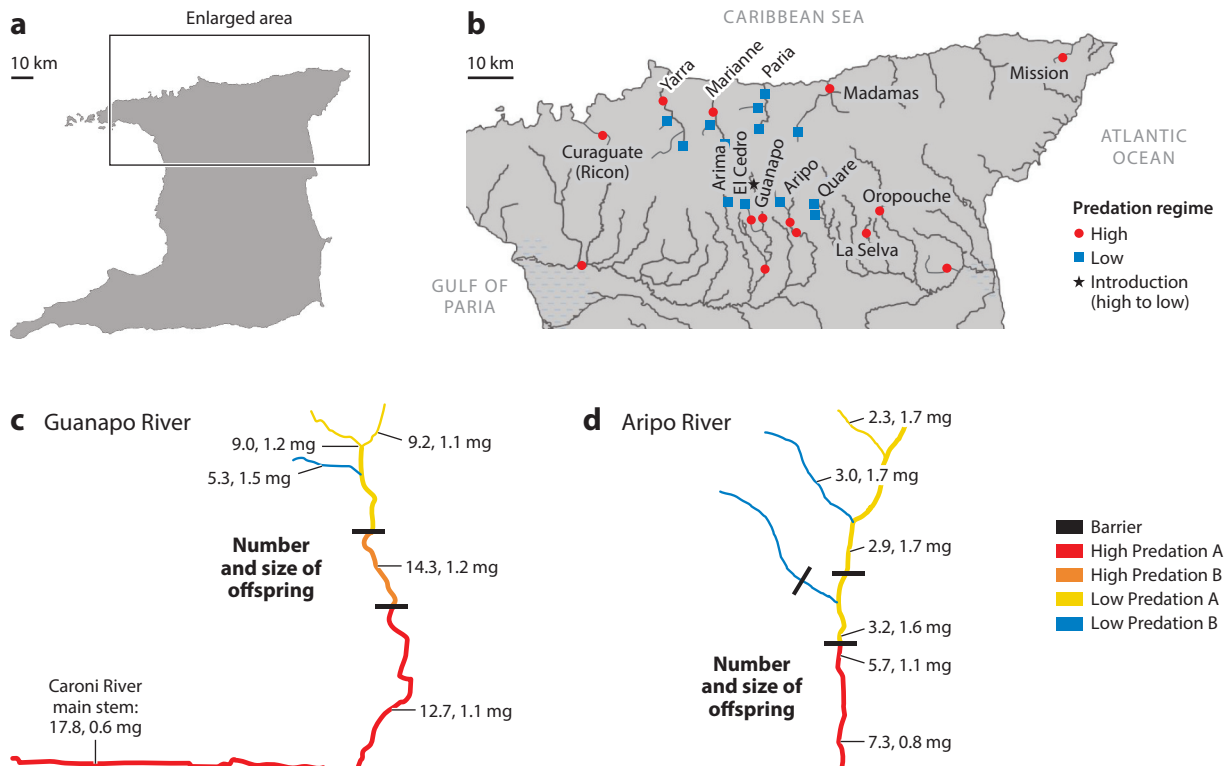


Figure 1

(a) Map of Trinidad. (b) Details of a subset of the rivers and high- and low-predation localities included in our studies. The star on the El Cedro River represents one of our introduction sites, where guppies were transplanted from a high-predation site below a barrier waterfall into a guppy-free, low-predation site above a barrier waterfall. (c,d) Details of the distribution of life history phenotypes in two rivers illustrate the full diversity of community types. In most work, we contrast high-predation and low-predation communities, but some intermediate communities contain guppies with intermediate phenotypes. The first of each pair of numbers is the least square mean reproductive allocation, or the percent of total dry mass that consists of developing embryos. The second number is the mean dry mass (mg) of developing embryos. (c) The Guanapo River. The barrier separates High Predation A (red) from High Predation B (orange). The waterfall excludes *Crenicichla alta* but not other predators. The barrier between High Predation B and Low Predation A (yellow) excludes almost all remaining predators. Only *Rhamdia sebae* and *Rivulus bartii* are found above that barrier. Only *R. bartii* is found in Low Predation B communities (blue). (d) The Aripo River. The barrier that separates High Predation A from Low Predation A excludes all fish except the cichlid *Aequidens pulcher*, *R. bartii*, and nonpredaceous species. Only *R. bartii* is found in Low Predation B communities (blue).

Trinidad guppies to any populations in mainland South America. The distinctness of the west versus east guppies defines the first two biogeographic regions. The third biogeographic region is the set of streams draining the north slope, which appear never to have had a mainland connection. Alexander et al. (2006), on the basis of the base pair sequences of Xsrc (a nuclear locus), inferred that the guppies from the western tributaries of the north slope (Yarra, Marianne, and Paria Rivers) are most closely allied to those from the Caroni drainage and the Orinoco Basin. Guppies from the one river sampled on the east side of the north slope (Madamas River) instead clustered with the Oropouche drainage guppies. In contrast, Willing et al. (2010), on the basis of a genome-wide SNP scan, and Suk & Neff (2009), on the basis of microsatellite data, inferred that the three biogeographic regions are distinct from each other. The genetic diversity of guppies south of the Northern Range Mountains has not been well explored. The island may well host other distinct clades.

The guppies from each major tributary within each of the three biogeographic regions are also genetically distinct from one another. Paired populations from high- and low-predation localities within each tributary are genetically more similar to each other than they are to fish from other, even neighboring, tributaries. This genetic similarity belies the substantial phenotypic differences between guppies from high- and low-predation communities within each river (Alexander et al. 2006, Fajen & Breden 1992, Fraser et al. 2015, Suk & Neff 2009, Willing et al. 2010). Exceptions to this pattern are sometimes seen in the genetic similarities among high-predation communities in the lower reaches of neighboring rivers, suggesting that ongoing gene exchange occurs between them. This pattern of variation argues that virtually all rivers in which we have compared the phenotypes of guppies from high- and low-predation communities represent independent instances in which guppies have adapted to life with and without predators.

2.4. Theoretical Framework

Early investigators (Haskins et al. 1961, Seghers 1973) built a circumstantial case for differences between high- and low-predation communities in size- and hence age-specific predation. *Crenicichla alta*, a dominant predator in high-predation streams, feeds preferentially on large, adult size classes of guppies. *R. hartii*, the only predatory fish in low-predation streams, feeds predominantly on small, immature size classes of guppies. Endler (1978, 1980) argued that the absence of visually oriented predators found in high-predation communities allowed males in low-predation communities to evolve more colorful patterns.

In the 1970s, when this research program began, a few theoretical papers predicted how life histories would evolve in response to changes in age-specific mortality risk. Early, density-independent theories of the vintage of Gadgil & Bossert (1970), Schaffer (1974b), Law (1979), Michod (1979), and Charlesworth (1980) all predicted that high adult mortality risk would select for earlier maturity and a higher rate of investment in reproduction, with the opposite expected in environments with selective predation on juveniles.

Theory that included density-dependent regulation made similar predictions, provided that there was heterogeneity among age classes in their susceptibility to the deleterious effects of high population density (Charlesworth 1980). The theory of *r*- and *K*-selection was more prominent but seemed less relevant because it excluded age-specific differences in mortality risk. Theoretical treatments also addressed the role of fluctuating environmental conditions on the evolution of life histories (Murphy 1968, Schaffer 1974a). This body of work was not central to our research at the time, partly because testing this theory would require long-term data and partly because we knew nothing about environmental variation in this system.

Testing these three flavors of life history theory has one major challenge: Certain patterns of life history can be, on the surface, compatible with all of them. This means that merely quantifying life history and ecological differences would not be sufficient, and a variety of approaches and experiments would be necessary to falsify any one of them. In recent years, these theoretical perspectives have been unified (Engen & Sæther 2016, Engen et al. 2013, Lande et al. 2009), and novel methods for bringing empirical data to bear on theory have been suggested. In addition, as we show, we have been collecting the long-term data necessary for rigorously testing this entire body of theory.

3. METHODS

Our research is a combination of retrospective and prospective studies of evolution. Retrospective studies quantify the differences among guppies adapted to high- and low-predation communities;

they reveal the footprint of adaptive evolution. Prospective studies are experimental studies in which we look at evolution in action. Combining the two allows us to test a variety of hypotheses about how and why guppies have evolved.

The elements of our retrospective research include the following:

1. Comparative studies of natural populations in which we characterize life histories and the comparative ecology of guppies from high- and low-predation environments. For examples, see Reznick (1989), Reznick & Endler (1982), and Reznick et al. (1996a).
2. Laboratory common garden experiments in which we characterize differences among populations in life histories of the laboratory-reared grandchildren of wild-caught adults. Rearing fish through two generations under controlled laboratory conditions effectively removes confounding environmental influences on the phenotypes of guppies collected from high-versus low-predation localities. Differences that persist are assumed to have a genetic basis (e.g., see Reznick 1982b, Reznick & Bryga 1996).
3. Mark-recapture studies that characterize mortality, growth rate, and population density in high- and low-predation environments. This work was coupled with a quantification of the comparative ecology of high- and low-predation environments (Reznick et al. 2001) (e.g., see Bryant & Reznick 2004, Reznick & Bryant 2007, Reznick et al. 1996a).
4. In situ manipulations that have enabled us to demonstrate density regulation in action in low-predation streams plus characterize ecological interactions between guppies and killifish. For examples, see Bassar et al. (2013) and Reznick et al. (2012).
5. Factorial experiments in artificial streams. Here we compared the performance of guppies derived from natural high- and low-predation environments. We characterized the performance (e.g., response to density, impact on the ecosystem) of high- and low-predation guppies in a common environment that replicated a natural stream. Our comparative ecology study (Reznick et al. 2001) enabled us to parameterize the density and size distributions of guppies to match those typical of natural high- and low-predation populations. This earlier work also provided a frame of reference for evaluating whether the growth rates, mortality rates, and fecundity correspond to values seen in natural streams. They did. For examples, see Bassar et al. (2010, 2012, 2013, 2017a,b).

Our prospective studies are the experimental transplantation of guppies derived from high-predation environments over barrier waterfalls that excluded guppies and predators. The only fish species found above these falls is the killifish *R. bartii*. Two such experiments were initiated last century (Aripo River, 1976, by John Endler; El Cedro River, 1981, by D. Reznick). A new generation of four replicate introductions was initiated in 2008 and 2009 by us. In all of these experiments, fish from the ancestral high-predation locality serve as controls for evaluating evolution in the introduced populations. In our new experiments, still ongoing, we collected guppies as juveniles from a high-predation site, reared them to maturity in single-sex groups, then mated them before introducing them into the four streams. Initiating the experiment this way means that we know all founders. All introduced fish were individually marked with subcutaneous injections of colorized elastomer (Northwest Marine Technologies). All populations are censused every month, and all new recruits are marked. We save scales from every individual to serve as a source of DNA. For sample publications derived from these experiments, see Auer et al. (2012), Kemp et al. (2018), and Reznick et al. (1990, 1997, 2019a).

The design of our introduction experiments facilitates two additional categories of studies. First, we are able to construct the pedigrees from the DNA in the scales kept from every guppy in all four streams. Thus far, we have only done so for one of the four streams because of funding

limitations. The pedigrees are combined with monthly census data to yield estimates of individual reproductive success, heritability of life history traits, and diverse other features of individual performance. Second, we are also able to construct before-after, control, and introduction studies of diverse phenomena because we initiated monthly assessments of invertebrate abundance and *aufwuchs* and bimonthly mark-recapture studies of killifish in the introduction sites and control region just upstream of the introduction sites, above natural barriers that exclude the introduced populations of guppies. These data provide the background for assessing the impact of guppies on their ecosystem, to be integrated with analyses of guppy evolution. See Simon et al. (2017) and Reznick et al. (2019a) for examples of all of these applications.

4. WHAT WE HAVE LEARNED FROM GUPPIES

4.1. Retrospective Evolution

Pronounced and consistent differences are apparent in the life histories of low-predation and high-predation guppies. High-predation guppies are smaller and younger at sexual maturity. They produce more offspring per litter, and individual offspring size is smaller. Combined, these factors make the reproductive biomass per litter (number of offspring $\times$ offspring size) larger. The time interval between successive broods is shorter. The combination of initiating reproduction earlier, reproducing more often, and putting more resources into each litter means that the rate of investment in reproduction is much higher in guppies from high-predation environments. These differences in the life history phenotypes of wild-caught fish have been replicated in different rivers from all three biogeographic regions in the Northern Range (Reznick 1982b, 1989; Reznick & Bryga 1996; Reznick & Endler 1982; Reznick et al. 1996b) (Figure 1b,c). These differences persist in the grandchildren of wild-caught fish when reared in the laboratory, one per aquarium, on the same amount of food.

Multiple hypotheses are consistent with these results (Reznick & Bryga 1987, Reznick & Endler 1982). First, they could be a direct consequence of size-specific predation. Second, they could be an indirect consequence of the differences in predation. High or low predation rates produce low or high population densities of guppies, respectively. The life histories by themselves cannot distinguish between these alternative hypotheses. When we discovered these life history differences (in the early 1980s), little was known about either the real differences in mortality rate, the magnitude of differences in density, the extent to which the populations were regulated, or anything else about the comparative ecology of guppies. These unknowns defined the next steps in our research.

4.2. Comparative Population Dynamics and Ecology

Our next steps involved developing methods for characterizing the comparative demography and ecology of natural populations of guppies from these alternative environments.

4.2.1. Population biology. We developed methods to individually mark guppies and then performed mark-recapture studies in several high- and low-predation populations. Mark-recapture studies showed that high-predation guppies experience substantially higher mortality rates than low-predation guppies (Reznick & Bryant 2007, Reznick et al. 1996a), consistent with the hypothesis that direct effects of predation select for the high-predation phenotype and alleviating these effects selects for the low-predation phenotype. However, for this to be so, theory tells us that adults in high-predation environments must be at greater risk than juveniles and the opposite pattern must occur in low-predation populations. In fact, we found that the differences in mortality

rates were equally distributed across all size classes. This result suggested that the direct effects of predation could not be causing these differences in life history.

4.2.2. Comparative ecology. We quantified features of the comparative ecology of high- and low-predation communities. High-predation guppies have lower population densities; are dominated by smaller, younger fish; and have higher individual somatic growth rates than low-predation populations (Reznick & Bryant 2007, Reznick et al. 1996a). The numerical dominance of small, young fish in high-predation populations is, in part, attributable to demography (high-predation guppies have higher birth and death rates). But ecological differences also contribute to these patterns: Predators reduce guppy population densities and increase per capita food availability to the survivors. In the absence of predators, guppy population density increases, leading to resource limitation and slower somatic growth. The difference in resource limitation is accentuated by the fact that low-predation environments are smaller streams with lower light availability. No other differences between the respective habitats could offer alternative explanations for higher growth rates in high-predation environments (Reznick et al. 2001).

Guppies in low-predation environments are resource limited and are adapted to life at high population densities. Low-predation guppies have smaller, asymptotic body sizes (Reznick & Bryant 2007) and produce larger offspring (Reznick 1982a,b), which is an adaptation to a food-limited environment (Bashey 2008, Jørgensen et al. 2011). They produce fewer offspring per litter in both the laboratory and the field. The magnitude of this difference in fecundity is much smaller in the laboratory, where low- and high-predation guppies are reared on the same food rations, than it is in the field (Reznick & Travis 1996, Reznick et al. 1996b). This pattern suggests that the field results combine a genetic predisposition of low-predation guppies to produce fewer offspring with lower food availability.

4.2.3. Density regulation. If the absence of large predators and the associated increase in guppy population density cause low-predation populations to become density regulated, increasing their densities in the field experimentally should produce demographic changes that would lead to a decline in population density. Reducing their density should have the opposite effect.

We exploited the pool–riffle structure of many natural streams to test this hypothesis by manipulating guppy density within pools in low-predation environments. For each replicate of the experiment, we selected three pools similar in structure and close to one another in a stream. We captured, measured, and marked all fish from the three pools and then reintroduced them at three different densities—typical density, reduced density, and increased density. We recollected them approximately 20 days later; quantified survival, growth, and reproduction; and then applied integral projection models (IPMs) to estimate the instantaneous population growth rates of the three treatments. The reduced-density treatments had accelerated population growth rates, and the increased-density treatments had decelerated population growth rates; the ambient-density treatments tended to remain stable in population size (Bassar et al. 2013, Reznick et al. 2012).

4.2.4. Density-dependent selection. We compared the responses of high- and low-predation guppies to high and low population densities in artificial streams (see Section 3 for methods). The experiment was similar to the manipulations of density in natural streams, in that marked and measured guppies were added to each channel and then recollected after 28 days. We then applied IPMs to the demographic data to estimate population growth rate, which is also a point estimate of mean fitness for that cohort. High-predation guppies had higher mean fitness than low-predation guppies in the low-density treatment but equal mean fitness at high population densities (Bassar et al. 2013). Both high- and low-predation guppies had reduced mean fitness at

high population densities, but the high-predation guppies were much more sensitive to density and hence sustained a greater loss of fitness than low-predation guppies.

The results for density regulation and density-dependent selection show that the indirect effects of predators, mediated through their impact on population density, play a role in shaping how guppies adapt to low-predation environments.



4.3. Senescence

Our interest in the evolution of senescence may seem a digression, but the results created a challenge that helped shape subsequent research. The low mortality rates experienced by guppies in low-predation environments are predicted to favor the evolution of delayed senescence, on the basis of G.C. Williams's antagonistic pleiotropy hypothesis for the evolution of senescence (Williams 1957). Williams hypothesized that the evolution of senescence is shaped by trade-offs between the contribution of reproduction to fitness early and late in life. When selection favors increased investment in reproduction early in life, the reduced investment in somatic maintenance to ensure reproduction later in life should cause some form of systemic breakdown that is independent of extrinsically imposed mortality, such as that by predators. However, high-predation guppies are unlikely to live long enough in nature to pay that price. Absent predation, low-predation guppies' life spans should be extended enough to realize the costs of high, early investment. Low-predation guppies' lower investment in reproduction and greater prospects of surviving to advanced ages led us to predict that the rate of deterioration with age should be higher in high-predation guppies than in low-predation guppies.

We quantified senescence as age-specific reproduction, mortality rate, and swimming performance in a common garden setting with second-generation, laboratory-born guppies from the low-predation environments of the Yarra River (north slope) and the Oropouche River (east Trinidad). Our results were qualitatively the same for both paired comparisons and were, for the most part, opposite to predictions. High-predation guppies were, as expected, younger at maturity, reproduced more often, and produced more babies per litter. However, in the laboratory, they had lower mortality rates, longer life spans, continued to reproduce at more advanced ages, and had a lower rate of decline in the rate of reproduction than their counterparts from low-predation environments (Reznick et al. 2004).

These results defy the trade-offs assumed in virtually all theoretical treatments of life history evolution. They also beg the question—why does the low-predation life history evolve? These results by themselves imply that the high-predation phenotype is unconditionally superior to the low-predation phenotype, so the low-predation phenotype should never evolve. Yet it does. Our finding that low-predation guppies are less sensitive to high population densities represents a partial answer. The senescence study was performed on fish kept one per aquarium and fed controlled rations, as in all of our common garden studies. The artificial stream experiment shows that high population densities erode the high-predation advantage, but density alone cannot explain the evolution of the low-predation phenotype. High densities cause the fitness of the two to be equal, but the low-predation guppies must be superior under some conditions if that phenotype is to evolve.

4.4. Guppy–*Rivulus* Interactions

Killifish are also distributed across the full spectrum of fish communities, but they exceed the range of guppies by invading even the smallest and least accessible mountain streams, where they are the only fish species present. Killifish life histories have also evolved in response to high- and low-predation communities in all three biogeographic regions (Furness & Reznick 2014;

Furness et al. 2012; Walsh & Reznick 2008, 2009, 2010; Walsh et al. 2011). Guppies and killifish are engaged in bidirectional, intraguild predation: The adults of each species prey on the juveniles of the other (Fraser & Lamphere 2013, Furness & Reznick 2014), and adult guppies compete with juvenile killifish (Gilliam et al. 1993). Guppies cause a 50% reduction in the population density of killifish and an increase in their per capita food availability. Killifish that co-occur with guppies grow faster and have higher rates of reproduction relative to those that occur without guppies.

Killifish in killifish-only communities are like guppies in low-predation communities, limited by intraspecific competition for food. When they are transplanted into the guppy–killifish community downstream, below the barrier waterfall that excludes guppies, their growth rates accelerate to match those of resident killifish (Walsh et al. 2011). Common garden laboratory studies show that killifish adapted to life with guppies are younger at maturity and have higher rates of egg production than those adapted to streams in which they are the only species present (Walsh & Reznick 2010). Like guppies, these adaptive differences can be seen in populations found just above and just below the barrier waterfalls that exclude guppies.

The coadaptation of guppies and killifish plays a significant role in defining the conditions that favor the evolution of the low-predation phenotype (Bassar et al. 2017b). The design of this experiment mimics the presumed sequence of events in which high-predation guppies invade, and adapt to, a stream previously occupied by killifish alone: (a) invasion, high-predation guppies paired with *Rivulus* from a killifish-only location; (b) guppy adaptation, low-predation guppies paired with killifish from a killifish-only location; and (c) killifish adaptation, low-predation guppies paired with killifish from a killifish–guppy environment. As in all such experiments, this one was replicated by running it first with guppies and killifish from the three habitat types in one river (the Aripo River, west Trinidad) and then again with fish from all three habitat types in a different river (the Quare River, east Trinidad).

The low-predation guppies had higher population growth rates, and hence higher mean fitness, than high-predation guppies when both were paired with killifish from a *Rivulus*-only site. The presence of killifish is thus a necessary component of the conditions that make low-predation guppies superior to high-predation guppies. When low-predation guppies were paired with killifish from a low-predation site, their fitness was higher than that of low-predation guppies paired with killifish from a killifish-only site. This result shows that the coadaptation of killifish and guppies increases guppy fitness and helps shape the low-predation phenotype.

4.5. Prospective Evolution: Initial Experiments

We have exploited the presence of barrier waterfalls to perform experimental studies of evolution in natural populations of guppies. We have manipulated the mortality risks of guppies by transplanting them from high-predation localities below barrier waterfalls into previously guppy-free localities above barrier waterfalls. Introduced guppies evolved delayed maturation and reduced reproductive allocation, as seen in natural low-predation communities.

In the first pair of experiments, initiated in 1976 and 1981, we found that male age and size at maturity evolved to the values typical of low-predation communities within 4 years (Endler 1980; Reznick & Bryga 1987; Reznick et al. 1990, 1997). Female life history traits evolved as predicted in 11 years. Other attributes of guppies, including behavior (O'Steen et al. 2002), also evolved. This rate of evolution is striking. Stearns (1992) showed that the rates of guppy life history evolution were four to seven orders of magnitude faster than those observed in the fossil record and were of the same order of magnitude as seen in artificial selection. This high rate of contemporary evolution was a novelty when we first reported it in 1987 (Reznick & Bryga 1987) and 1990 (Reznick et al. 1990).

4.6. Prospective Evolution: Eco-Evo Dynamics

The second set of experiments reflects our interest in the consequences of contemporary evolution. We were attracted to the concept of interactions between ecology and contemporary evolution, hereafter referred to as eco-evo dynamics, which was initially conceived by David Pimentel (1961) but was brought to our attention by the outstanding series of papers by Hairston, Ellner, and colleagues on predator–prey dynamics in a model ecosystem (Fussmann et al. 2003, Shertzer et al. 2002, Yoshida et al. 2004). Pimentel (1961, p. 65) framed his hypothesis as applied to density-dependent evolution: “Density influences selection; selection influences genetic make-up; and in turn, genetic make-up influences density.” This hypothesis was a natural fit for the guppy system, and we had the foundation upon which we could test it: ecological differences between high- and low-predation environments, density regulation in low-predation environments, evidence for density-dependent selection, and demonstrably rapid evolution.

To test whether this kind of feedback characterizes the guppy system, we addressed three questions. First, do guppies affect their ecosystem in a manner that alters the selective pressures they encounter? Second, do guppies respond to those altered selective pressures? Third, does guppy evolution, in turn, affect aspects of the ecosystem and might those changes cause further evolution in guppies?

The new prospective study—four replicated transplantations of high-predation guppies into four guppy-free streams—was designed to ask if and how such interactions unfold in a natural ecosystem. New elements of these introductions include beginning with wild-caught juveniles reared to adulthood in single-sex groups that were then mated, marked, and introduced. In prior work, we introduced wild-caught adults, which included females carrying stored sperm, which means we did not know all founders. In the new study, we keep scales from all founders to provide DNA for individual recognition. Introduced guppies are censused monthly. All new recruits are individually marked and scales are saved. Our detection probabilities in any given month exceed 90%. We have a greater than 99% probability of seeing every fish recruited to the adult population. The successive monthly surveys reveal changes in population density and enable us to follow the fates of individuals and measure aspects of their life history. The DNA samples enable us to construct pedigrees and then estimate quantitative genetic parameters and lifetime reproductive success. We perform an annual common garden assessment of life histories in the four introduced populations and the ancestral high-predation population.

We preceded these new introductions with one year of data collection in the introduction site and a control site upstream in all four streams, isolated from the guppy introductions by barrier waterfalls. We initiated mark-recapture censuses of resident killifish and assessments of the ecosystem, including quantification of the abundance of benthic invertebrates, algae, and detritus. These data characterize how guppy populations change over time in concert with correlated changes in killifish and other features of the ecosystem. This time series is complemented with short-term experiments in artificial and natural streams designed to decipher cause-and-effect relationships that underlie the events unfolding in the introduction experiments. This work is ongoing; here, we present highlights.

4.6.1. Do guppies affect their ecosystem in a manner that alters selection on guppies?

The introduced guppy populations grew to peak densities in two to three years and then fluctuated seasonally. Populations increase during the dry season and decrease during the wet season, when floods scour the streams of benthic organisms, causing food availability to decline (Reznick et al. 2019a). Within the first 18 months of the guppy introduction, the abundance of juvenile killifish declined, causing a significant increase in the mean size of killifish in the portion of stream where guppies had been introduced, relative to the guppy-free control reach upstream

(Fraser & Lamphere 2013). Killifish abundance has continued to decline in the guppy introduction sites relative to the upstream controls in all four streams (J. Goldberg, personal communication). Guppies significantly depleted invertebrate abundance and diversity [T. Heatherly & S. Thompson, personal communication, described in Travis et al. (2014)]. However, this decline in invertebrate abundance was not evident until two years into the study, when guppies had attained peak population densities.

We complemented this work with a mesocosm experiment (Bassar et al. 2010, 2012) in which we stocked artificial streams with guppies derived from natural high- or low-predation environments at high or low population densities. We quantified and compared the ecosystem impacts of guppies and evaluated diet choice via stomach content analyses of all fish at the end of the experiment.

All guppy populations depleted the environment of benthic invertebrates and algae and did so to a greater degree when they were at higher population densities. They indirectly increased primary productivity, because their grazing activities kept the substrate clear of sediment. Experiments in natural streams, in which we excluded guppies from quadrats, proved that guppies have comparable impacts on algae and invertebrates in natural streams (Simon et al. 2017).

Guppies alter their ecosystem, but do those changes alter the selection they experience? Further results from this experiment suggest that the impact of guppies can indeed cause guppy evolution. The high-predation and low-predation guppies differed in both their direct and indirect effects on the ecosystem because of their diets. High-predation guppies selectively depleted the environment of high-quality invertebrates, meaning invertebrate prey that have low C:N ratios, or higher protein content. Their algal consumption was also selective: Guppies harvested the dominant diatom species and hence changed the algal community (Bassar et al. 2017a). By contrast, low-predation guppies consumed all benthos (algae, detritus, invertebrates) in proportion to each one's relative abundance.

The results suggest that the change in diet is a response to the guppy-induced change in resource availability. High-predation guppies are adapted to environments with low guppy population densities and abundant resources; hence, they specialize on high-quality resources. When introduced to low-predation environments, they first deplete the environment of high-quality prey and then become more generalist consumers. This result is complemented by models showing that the effect of high-predation guppies on the ecosystem generates selection favoring greater herbivory, which is precisely what we see in the diet of low-predation guppies (Bassar et al. 2012).

4.6.2. Do guppies evolve in response to their impact on the ecosystem? The temporal patterns of guppy evolution in our introduction experiment, combined with patterns of population and community dynamics, can help us discriminate between the roles of the direct effect of reduced predation (size-specific predation) and the indirect effect of reduced predation (density-dependent evolution) in shaping guppy evolution. If direct effects dominate, selection should be strongest and evolution fastest when guppies are first introduced, because that is when the difference between their phenotypes and the optimal phenotype is greatest. As they evolve toward a new adaptive peak, this difference will decline. The magnitude of the selection coefficient and the rate of evolution should decline in proportion to this difference. If indirect, density-mediated effects dominate, selection will initially be absent because populations are introduced at low densities and take two to three years to grow to peak densities. The intensity of selection should increase over time as the growing population of guppies impacts the structure of the ecosystem and changes the optimal phenotype, thereby creating a difference between the average phenotype and the new optimum. These hypotheses are not mutually exclusive. Indeed, direct selection could dominate and still produce a lag in the evolutionary response. We discuss these complications elsewhere (Reznick et al. 2019a).

Male age and size at maturity evolves primarily as an indirect response, mediated by the emergence of density regulation (Reznick et al. 2019a). A two- to three-year lag occurred between when guppies were introduced, when they attained peak population densities, and when we observed a significant increase in age and size at maturity in the introduced versus ancestral populations in our annual common garden assessment of life histories. This evolution was accompanied by declining individual growth rates and declining lifetime reproductive success in the residents of the introduction sites; these results imply the emergence of density regulation. At the same time, a signal of density-dependent regulation emerged in three of four streams in the form of a negative correlation between the rates of density changes in successive months: Increases tended to be followed by decreases, and vice versa (J. Travis, R. Bassar, D. Reznick & T. Coulson, manuscript in preparation).

By contrast, male coloration appears to evolve as a direct consequence of a reduced risk of predation. Endler (1980) observed significant increases in the number and size of color spots of males when he first evaluated his introduced population, two years after their transplantation from a high-predation locality into a previously guppy-free, low-predation locality. We recorded significant evolution of male coloration in our first annual common garden assessment, initiated with juveniles collected 12 months after the introduction (Gordon et al. 2015, Kemp et al. 2018). These rapid, significant responses are consistent with there being a direct effect of predators and female preference on male coloration.

4.6.3. Does guppy evolution, in turn, affect aspects of the ecosystem? At the present time, our answer to this question is based on retrospective analyses. The strikingly different effects of high-predation and low-predation guppies on their ecosystems, as revealed in the mesocosm study, indicate that guppy evolution affects the ecology of their communities. Perhaps the most striking aspect of this effect is the large difference in indirect effects, which follows from their dietary differences, differences in excretion rates (El-Sabaawi et al. 2015), and differences in biomass distributions of guppies in low-predation and high-predation populations (Bassar et al. 2012).

Simon et al. (2017) compared sites with guppies with sites just upstream of a waterfall that excluded guppies. In four natural low-predation streams occupied by locally adapted guppies, guppies reduced algal but not invertebrate abundance. In our introduction sites, occupied by recently introduced guppies from high-predation streams, guppies instead reduced the abundance of invertebrates. These results were echoed in a landscape-scale survey of several watersheds in the Northern Range (R. El-Sabaawi, personal communication).

Given these results, we expect to see long-term changes in our introduction streams as guppies continue to evolve. We should see the impact of our introduced guppies shift over time to correspond to and to match the impact of guppies in natural low-predation environments as the guppies adapt to their new, guppy-modified ecosystem.

5. WHAT GUPPIES CAN TEACH EVERYONE

We started this work, and this article, by addressing the evolution of life histories. Pursuing this interest led to discoveries that offered answers to the following, more general questions in evolutionary biology addressed in our introduction.

5.1. How Fast and How Repeatable Is Evolution?

Descriptions of contemporary evolution in other systems (Hendry & Kinnison 1999, Reznick & Ghalambor 2001) demonstrate that observing it in guppies was no fluke. Of course, contemporary

evolution had been documented long before we began our work [e.g., *Myxoma* (Pimentel 1961), heavy metal tolerance in plants (McNeilly & Antonovics 1968), industrial melanism (Kettlewell 1973)]. However, the earliest demonstrations of contemporary evolution were responses to strong anthropogenic selection pressures. The understanding that evolution in response to natural selection could occur just as rapidly was a novel result.

The patterns of evolution we and others have seen in guppies have proven highly repeatable across the three biogeographic regions of Trinidad and across two very different types of fish communities—one dominated by species typical of South American freshwater streams, the other by oceanic species. The rate of evolution has also proven repeatable in the two original and four new introduction experiments.

5.2. Does the Speed of Evolution Inform Us About the Underlying Genetics?

The rapid and repeatable evolution of life histories in six introduced populations means that this evolution was fueled by standing genetic variation rather than by new mutations. The finding that natural populations harbor extensive standing genetic variation is not a novel discovery. What is striking is the repeated demonstration of rapid adaptation, in the same manner, to a novel environment. On the one hand, this result supports the well-known hypotheses that much of the standing variation in populations can be adaptive (Lewontin 1974). On the other hand, our results focus attention on the fundamental paradox of evolutionary biology—namely, how genetic variation is preserved (Mitchell-Olds et al. 2007).

One potent mechanism for preserving variation, demonstrated in natural populations of guppies, is frequency-dependent selection. Experiments in natural high- and low-predation environments proved that rare male color phenotypes have higher survival and higher mating success than more common phenotypes (Hughes et al. 2013, Olendorf et al. 2006). These articles provide only a partial answer to how heritable variation is preserved, as much more than male coloration predictably evolves as guppies adapt to life with and without predators.

5.3. Are the Adaptations We See in Natural Populations Driven by a Single, Dominant Agent of Selection or by Multiple, Potentially Interacting Agents of Selection?

An ecologist's perspective and prior research on other organisms (Jones et al. 1977) make it seem obvious that adaptation in nature is likely to be driven by multiple factors (Reznick & Travis 1996). The more interesting questions are which agents of selection act on which traits, how do they combine to drive adaptive change, and how do the relationships among traits shape the responses to selection.

Guppy adaptation to the different fish communities is clearly driven by multiple, interacting agents of selection. Predation, the interaction between predation and guppy population density, and the interaction between predation and sexual selection all drive the evolution of the alternative phenotypes. Predation likely acts directly on morphology and locomotory ability, but the evolution of some aspects of locomotion and life history interact (Ghalambor et al. 2004). The interaction between predation and guppy population density embraces several ecological factors that can act cumulatively on foraging, diet, assimilation efficiency, physiology, trophic morphology, and resource allocation through the cascading impacts of guppies on their ecosystem (Auer et al. 2018, Palkovacs et al. 2011, Reznick et al. 2004, Zandonata et al. 2011): the depletion of high-quality invertebrate prey, the selective removal of the competitively dominant diatom (Bassar et al. 2010, 2017a), and the effects of crowding on pathogen transmission and social stress. The interaction between predation and sexual selection acts on male color pattern and behavior

(Endler 1978, Endler & Houde 1995). In addition, the complex interaction between guppies and killifish in low-predation environments plays an unexpectedly important role in shaping the low-predation phenotype (Bassar et al. 2017b).

The relationships among individual traits, be they trade-offs (negative association) or syndromes (positive association), also shape adaptation. We began this project by focusing on life history theory and the costs of reproduction as the major trade-offs shaping adaptive change. Our comparative study of senescence forced us to abandon this simple framework and ask a larger question: What are the critical features of the low-predation phenotype, other than life history, and why did the low-predation phenotype evolve? This led to a consideration of interactions between patterns in resource allocation, which is a life history evolution problem, and the associated patterns of resource abundance and acquisition, which are ecological problems (Reznick et al. 2000). The cascading effects of high guppy population densities in low-predation environments have selected a suite of changes in diet, foraging, and efficiency of resource use, all of which confer increased interspecific competitive ability. At the same time, this ability offsets the absence of expected trade-offs among life history traits. In this light, it is natural to be drawn to Pimentel's (1961) suggestion that there can be a dynamic interaction between contemporary evolution and ecology. We have yet to understand all of the connections among trophic adaptations and life history evolution, but the trajectory of our work shows how unexpected results have provoked us to think more broadly about how the intrinsic properties of the whole organism affect its evolution in the real world.

5.4. How Much Ecology Do You Need to Understand Adaptive Evolution?

Few of our discoveries would have been possible without extensive, detailed ecological work. First, our comparative ecological studies helped define which features of the different communities in which guppies are found might be the agents of selection and which might be the consequences of guppy adaptation (Grether et al. 2001, Reznick et al. 2001). They also revealed new ecological distinctions between guppy phenotypes, including gut morphology and diet (Zandona et al. 2011) and stoichiometric ratios (El-Sabaawi et al. 2012). These studies also suggested the kinds of experiments we needed to do to unravel the underlying cause and effect.

Second, sound tests of life history theory require details of the autecology of the species in question, including demographic data. Life history theory is couched in terms of both viability and fertility selection (Charlesworth 1980), which require studying individuals. Our mark-recapture studies have been critical not only for their ecological value (e.g., numerical dynamics, diagnosing the onset of population regulation after introductions) but also for enabling us to recognize the importance of indirect effects in shaping life history evolution. When combined with our pedigree analyses in our ongoing introduction experiments, they are enabling us to incorporate the influence of seasonal and spatial variation in the selection guppies experience on shaping the attributes of the population as a whole (Auer et al. 2012; Bassar et al. 2013; Reznick & Bryant 2007; Reznick et al. 1996a, 2012).

5.5. Conversely, Do We Need to Understand Contemporary Evolution to Understand the Ecological Relationships Among Species in a Community?

Our work demonstrates the extensive reach of interactions between contemporary evolution and ecology in the form of the scope of guppy impacts on their ecosystem (Bassar et al. 2010) and the dramatic differences between the ecosystem impacts of guppies adapted to high- versus low-predation communities (Bassar et al. 2010, Simon et al. 2017). Even more remarkably, the different phenotypes exert different indirect effects on their ecosystems (Bassar et al. 2012), in this

case on nutrient cycling, mediated through the differences in their diets and through nitrogen and phosphorus excretion (El-Sabaawi et al. 2015). Early empirical work on interactions between ecology and evolution was based on simple two-species interactions in model ecosystems, leading to an understandable skepticism about whether such feedbacks could play any role in a more complex community. Our work shows that these interactions do indeed have strong effects on natural ecosystems.

5.6. Finally, Can We Understand Evolution by Simply Studying the Footprints of Adaptation in Living Organisms, or Is Something to Be Gained by Also Considering the Dynamics of the Process?

Our ability to observe evolutionary dynamics in this system, as opposed to earlier experiments that documented beginning and end points with comparisons among high- and low-predation guppies, has shown how the climb up the adaptive peak is, in fact, a struggle, not a simple hike. Our ongoing quantification of numerical dynamics and selection gradients has shown that these gradients vary seasonally and, for a given season, even among years. The trajectory of phenotypes experiences accelerations and decelerations as it ascends the peak, along with reversals and stasis. We have also found that males and females evolve at different rates (Reznick et al. 1997), setting the stage for an interesting sexual conflict as both approach their respective peaks. None of this could have been discovered merely by comparison.

Looking forward, we are using our ability to follow the dynamics of evolution to answer new questions in ecology and evolutionary biology. On the ecological side, we are asking how a bidirectional intraguild predation interaction can be stable. In addition, our ability to collect long-term data on population dynamics of interacting species is allowing us to evaluate the contribution of storage effects to species coexistence. On the evolutionary side, our mark-recapture studies in four replicate introduction experiments will create four pedigrees and a DNA archive of almost every individual in these populations. We can combine this pedigree with measures of performance and fitness to study the genetics of adaptation as it unfolds. We want to add guppies to the roster of populations whose long-term study has informed so much of evolution and ecology: the medium ground finch, great tits, red deer, and Soay sheep. We hope to use the added power of replication and experimental manipulation to explore issues these other systems could not address.

DISCLOSURE STATEMENT

The authors are not aware of any affiliations, memberships, funding, or financial holdings that might be perceived as affecting the objectivity of this review.

ACKNOWLEDGMENTS

The undergraduates, interns, and colleagues who contributed to this work are too many to name here; we could not have succeeded without them. We gratefully acknowledge the continuing support of the National Science Foundation: DEB 80-17011, DEB 82-14655, BSR 8620463, BSR-8818071, DEB-9119432, DEB-9419832, DEB-9707473, DEB-9801492, DEB-9968601, IBN-0111023, DEB-0623632EF, DEB-0808039, DEB-1258231, and DEB-1556884.

LITERATURE CITED

Alexander HJ, Taylor JS, Wu SST, Breden F. 2006. Parallel evolution and vicariance in the guppy (*Poecilia reticulata*) over multiple spatial and temporal scales. *Evolution* 60:2352–69

- Auer SK, Dick CA, Metcalfe NB, Reznick DN. 2018. Metabolic rate evolves rapidly and in parallel with the pace of life history. *Nat. Commun.* 9:14
- Auer SK, López-Sepulcre A, Heatherly T II, Kohler TJ, Bassar RD, et al. 2012. Life histories have a history: effects of past and present conditions on adult somatic growth rates in wild Trinidadian guppies. *J. Anim. Ecol.* 81:818–26
- Barson NJ, Cable J, van Oosterhout C. 2009. Population genetic analysis of microsatellite variation of guppies (*Poecilia reticulata*) in Trinidad and Tobago: evidence for a dynamic source–sink metapopulation structure, founder events and population bottlenecks. *J. Evol. Biol.* 22:485–97
- Bashey F. 2008. Competition as a selective mechanism for larger offspring size in guppies. *Oikos* 117:104–13
- Bassar RD, Bryan BL, Marshall MC, Pringle CM, Reznick DN, Travis J. 2017a. Local adaptation of fish consumers alters primary production through changes in algal community composition and diversity. *Oikos* 126:594–603
- Bassar RD, Ferriere R, López-Sepulcre A, Marshall MC, Travis J, et al. 2012. Direct and indirect ecosystem effects of evolutionary adaptation in the Trinidadian guppy (*Poecilia reticulata*). *Am. Nat.* 180:167–85
- Bassar RD, López-Sepulcre A, Reznick DN, Travis J. 2013. Experimental evidence for density-dependent regulation and selection on Trinidadian guppy life histories. *Am. Nat.* 181:25–38
- Bassar RD, Marshall MC, López-Sepulcre A, Zandon E, Auer SK, et al. 2010. Local adaptation in Trinidadian guppies alters ecosystem processes. *PNAS* 107:3616–21
- Bassar RD, Simon T, Roberts W, Travis J, Reznick DN. 2017b. The evolution of coexistence: Reciprocal adaptation promotes the assembly of a simple community. *Evolution* 71:373–85
- Bryant MJ, Reznick D. 2004. Comparative studies of senescence in natural populations of guppies. *Am. Nat.* 163:55–68
- Carvalho GR, Shaw PW, Magurran AE, Seghers BH. 1991. Marked genetic divergence revealed by allozymes among populations of the guppy *Poecilia reticulata* (Poeciliidae), in Trinidad. *Biol. J. Linn. Soc.* 42:389–405
- Charlesworth B. 1980. *Evolution in Age-Structured Populations*. Cambridge, UK: Cambridge Univ. Press
- Crispo E, Bentzen P, Reznick DN, Kinnison MT, Hendry AP. 2006. The relative influence of natural selection and geography on gene flow in guppies. *Mol. Ecol.* 15:49–62
- El-Sabaawi RW, Marshall MC, Bassar RD, López-Sepulcre A, Palkovacs EP, Dalton C. 2015. Assessing the effects of guppy life history evolution on nutrient recycling: from experiments to the field. *Freshw. Biol.* 60:590–601
- El-Sabaawi RW, Zandon E, Kohler TJ, Marshall MC, Moslemi JM, et al. 2012. Widespread intraspecific organismal stoichiometry among populations of the Trinidadian guppy. *Funct. Ecol.* 26:666–76
- Endler JA. 1978. A predator's view of animal color patterns. *Evol. Biol.* 11:319–64
- Endler JA. 1980. Natural selection on color patterns in *Poecilia reticulata*. *Evolution* 34:76–91
- Endler JA, Houde AE. 1995. Geographic variation in female preferences for male traits in *Poecilia reticulata*. *Evolution* 49:456–68
- Engen S, Lande R, Sæther BE. 2013. A quantitative genetic model of r - and K -selection in a fluctuating population. *Am. Nat.* 181:725–36
- Engen S, Sæther BE. 2016. Optimal age of maturity in fluctuating environments under r - and K -selection. *Oikos* 125:1577–85
- Fajen A, Breden F. 1992. Mitochondrial DNA sequence variation among natural populations of the Trinidad guppy, *Poecilia reticulata*. *Evolution* 46:1457–65
- Fraser BA, Künstner A, Reznick DN, Dreyer C, Weigel D. 2015. Population genomics of natural and experimental populations of guppies (*Poecilia reticulata*). *Mol. Ecol.* 24:389–408
- Fraser DF, Lamphere BA. 2013. Experimental evaluation of predation as a facilitator of invasion success in a stream fish. *Ecology* 94:640–49
- Furness AI, Reznick DN. 2014. The comparative ecology of a killifish (*Rivulus hartii*) across aquatic communities differing in predation intensity. *Evol. Ecol. Res.* 16:249–65
- Furness AI, Walsh MR, Reznick DN. 2012. Convergence of life-history phenotypes in a Trinidadian killifish (*Rivulus hartii*). *Evolution* 66:1240–54
- Fussmann GF, Ellner SP, Hairston NG. 2003. Evolution as a critical component of plankton dynamics. *Proc. R. Soc. B Biol. Sci.* 270:1015–22

- Gadgil M, Bossert WH. 1970. Life historical consequences of natural selection. *Am. Nat.* 104:1–24
- Ghalambor CK, Reznick DN, Walker JA. 2004. Constraints on adaptive evolution: the functional trade-off between reproduction and fast-start swimming performance in the Trinidadian guppy (*Poecilia reticulata*). *Am. Nat.* 164:38–50
- Gilliam JF, Fraser DF, Alkins-Koo M. 1993. Structure of a tropical fish community: a role for biotic interactions. *Ecology* 74:1856–70
- Gordon SP, Reznick D, Arendt JD, Roughton A, Hernandez MNO, et al. 2015. Selection analysis on the rapid evolution of a secondary sexual trait. *Proc. R. Soc. B Biol. Sci.* 282:20151244
- Grether GF, Millie DF, Bryant MJ, Reznick DN, Mayea W. 2001. Rain forest canopy cover, resource availability, and life history evolution in guppies. *Ecology* 82:1546–59
- Haskins CP, Haskins EG, McLaughlin JJA, Hewitt RE. 1961. Polymorphism and population structure in *Lebistes reticulatus*, a population study. In *Vertebrate Speciation*, ed. WF Blair, pp. 320–95. Austin: Univ. Tex. Press
- Hendry AP, Kinnison MT. 1999. The pace of modern life: measuring rates of contemporary microevolution. *Evolution* 53:1637–53
- Hughes KA, Houde AE, Price AC, Rodd FH. 2013. Mating advantage for rare males in wild guppy populations. *Nature* 503:108–10
- Jones JS, Leith BH, Rawlings P. 1977. Polymorphism in *Cepaea*: a problem with too many solutions? *Annu. Rev. Ecol. Syst.* 8:109–43
- Jørgensen C, Auer SK, Reznick DN. 2011. A model for optimal offspring size in fish, including live-bearing and parental effects. *Am. Nat.* 177:E119–35
- Kemp DJ, Batistic FK, Reznick DN. 2018. Predictable adaptive trajectories of sexual coloration in the wild: evidence from replicate experimental guppy populations. *Evolution* 72:2462–77
- Kettlewell B. 1973. *The Evolution of Melanism: The Study of a Recurring Necessity, with Special Reference to Industrial Melanism of the Lepidoptera*. Oxford, UK: Clarendon
- Lande R, Engen S, Sæther BE. 2009. An evolutionary maximum principle for density-dependent population dynamics in a fluctuating environment. *Philos. Trans. R. Soc. B Biol. Sci.* 364:1511–18
- Law R. 1979. Optimal life histories under age-specific predation. *Am. Nat.* 114:399–417
- Lewontin RC. 1974. *The Genetic Basis of Evolutionary Change*. New York: Columbia Univ. Press
- McNeilly T, Antonovics J. 1968. Evolution in closely adjacent plant populations. IV. Barriers to gene flow. *Heredity* 23:205–18
- Michod RE. 1979. Evolution of life histories in response to age-specific mortality factors. *Am. Nat.* 113:531–50
- Mitchell-Olds T, Willis JH, Goldstein DB. 2007. Which evolutionary processes influence natural genetic variation for phenotypic traits? *Nat. Rev. Genet.* 8:845–56
- Murphy GL. 1968. Pattern in life history and the environment. *Am. Nat.* 102:391–403
- Olendorf R, Rodd FH, Punzalan D, Houde AE, Hurt C, et al. 2006. Frequency-dependent survival in natural guppy populations. *Nature* 441:633–36
- O'Steen S, Cullum AJ, Bennett AF. 2002. Rapid evolution of escape ability in Trinidadian guppies (*Poecilia reticulata*). *Evolution* 56:776–84
- Palkovacs EP, Wasserman BA, Kinnison MT. 2011. Eco-evolutionary trophic dynamics: Loss of top predators drives trophic evolution and ecology of prey. *PLOS ONE* 6:e18879
- Pimentel D. 1961. Animal population regulation by genetic feedback mechanism. *Am. Nat.* 95:65–79
- Reznick DA, Bryga HA, Endler JA. 1990. Experimentally induced life-history evolution in a natural population. *Nature* 346:357–59
- Reznick DN. 1982a. Genetic determination of offspring size in the guppy (*Poecilia reticulata*). *Am. Nat.* 120:181–88
- Reznick DN. 1982b. The impact of predation on life history evolution in Trinidadian guppies: the genetic components of observed life history differences. *Evolution* 36:1236–50
- Reznick DN. 1989. Life history evolution in guppies. 2. Repeatability of field observations and the effects of season on life histories. *Evolution* 43:1285–97
- Reznick DN, Bassar RD, Handelsman CA, Ghalambor CK, Arendt J, et al. 2019a. Evo-evolutionary feedbacks predict the time course of rapid life history evolution. *Amer. Nat.* In press

- Reznick DN, Bassar RD, Travis J, Rodd FH. 2012. Life-history evolution in guppies VIII: the demographics of density regulation in guppies (*Poecilia reticulata*). *Evolution* 66:2903–15
- Reznick DN, Bryant MJ. 2007. Comparative long-term mark-recapture studies of guppies (*Poecilia reticulata*): differences among high and low predation localities in growth and survival. *Ann. Zool. Fenn.* 44:152–60
- Reznick DN, Bryant MJ, Roff D, Ghalambor CK, Ghalambor DE. 2004. Effect of extrinsic mortality on the evolution of senescence in guppies. *Nature* 431:1095–99
- Reznick DN, Bryga HA. 1987. Life-history evolution in guppies. 1. Phenotypic and genotypic changes in an introduction experiment. *Evolution* 41:1370–85
- Reznick DN, Bryga HA. 1996. Life-history evolution in guppies (*Poecilia reticulata*: Poeciliidae). V. Genetic basis of parallelism in life histories. *Am. Nat.* 147:339–59
- Reznick DN, Butler MJ IV, Rodd FH. 2001. Life history evolution in guppies. VII. The comparative ecology of high- and low-predation environments. *Am. Nat.* 157:126–40
- Reznick DN, Butler MJ IV, Rodd FH, Ross P. 1996a. Life-history evolution in guppies (*Poecilia reticulata*). 6. Differential mortality as a mechanism for natural selection. *Evolution* 50:1651–60
- Reznick DN, Endler JA. 1982. The impact of predation on life history evolution in Trinidadian guppies (*Poecilia reticulata*). *Evolution* 36:160–77
- Reznick DN, Ghalambor CK. 2001. The population ecology of contemporary adaptations: what empirical studies reveal about the conditions that promote adaptive evolution. *Genetica* 112:183–98
- Reznick DN, Losos JB, Travis J. 2019b. From low to high gear: There has been a paradigm shift in our understanding of evolution. *Ecol. Lett.* 22:233–44
- Reznick DN, Nunney L, Tessier A. 2000. Big houses, big cars, superfleas and the costs of reproduction. *Trends Ecol. Evol.* 15:421–25
- Reznick DN, Rodd FH, Cardenas M. 1996b. Life-history evolution in guppies (*Poecilia reticulata*: Poeciliidae). IV. Parallelism in life-history phenotypes. *Am. Nat.* 147:319–38
- Reznick DN, Shaw FH, Rodd FH, Shaw RG. 1997. Evaluation of the rate of evolution in natural populations of guppies (*Poecilia reticulata*). *Science* 275:1934–37
- Reznick DN, Travis J. 1996. The empirical study of adaptation in natural populations. In *Adaptation*, ed. MR Rose, GV Lauder, pp. 243–90. San Diego: Academic
- Schaffer WM. 1974a. Optimal reproductive effort in fluctuating environments. *Am. Nat.* 108:783–90
- Schaffer WM. 1974b. Selection for optimal life histories: the effects of age structure. *Ecology* 55:291–303
- Schories S, Meyer MK, Scharlt M. 2009. Description of *Poecilia (Acanthophaelus) obscura* n. sp., (Teleostei: Poeciliidae), a new guppy species from western Trinidad, with remarks on *P. wingei* and the status of the “Endler’s guppy.” *Zootaxa* 2266:35–50
- Seghers BH. 1973. *An analysis of geographic variation in the antipredator adaptations of the guppy, Poecilia reticulata*. PhD Thesis, Univ. B.C., Vancouver, Can.
- Shaw PW, Carvalho GR, Magurran AE, Seghers BH. 1991. Population differentiation in Trinidadian guppies (*Poecilia reticulata*): patterns and problems. *J. Fish Biol.* 39:203–9
- Shertzer KW, Ellner SP, Fussmann GF, Hairston NG. 2002. Predator-prey cycles in an aquatic microcosm: testing hypotheses of mechanism. *J. Anim. Ecol.* 71:802–15
- Simon TN, Bassar RD, Binderup AJ, Flecker AS, Freeman MC, et al. 2017. Local adaptation in Trinidadian guppies alters stream ecosystem structure at landscape scales despite high environmental variability. *Copeia* 105:504–13
- Stearns SC. 1992. *The Evolution of Life Histories*. Oxford, UK: Oxford Univ. Press
- Suk HY, Neff BD. 2009. Microsatellite genetic differentiation among populations of the Trinidadian guppy. *Heredity* 102:425–34
- Travis J, Reznick D, Bassar RD, López-Sepulcre A, Ferriere R, Coulson T. 2014. Do eco-evo feedbacks help us understand nature? Answers from studies of the Trinidadian guppy. *Adv. Ecol. Res.* 50:1–40
- Walsh MR, Fraser DF, Bassar RD, Reznick DN. 2011. The direct and indirect effects of guppies: implications for life-history evolution in *Rivulus hartii*. *Funct. Ecol.* 25:227–37
- Walsh MR, Reznick DN. 2008. Interactions between the direct and indirect effects of predators determine life history evolution in a killifish. *PNAS* 105:594–99

- Walsh MR, Reznick DN. 2009. Phenotypic diversification across an environmental gradient: a role for predators and resource availability on the evolution of life histories. *Evolution* 63:3201–13
- Walsh MR, Reznick DN. 2010. Influence of the indirect effects of guppies on life-history evolution in *Rivulus bartii*. *Evolution* 64:1583–93
- Williams GC. 1957. Pleiotropy, natural selection and the evolution of senescence. *Evolution* 11:398–411
- Willing EM, Bentzen P, van Oosterhout C, Hoffmann M, Cable J, et al. 2010. Genome-wide single nucleotide polymorphisms reveal population history and adaptive divergence in wild guppies. *Mol. Ecol.* 19:968–84
- Yoshida T, Hairston NG Jr., Ellner SP. 2004. Evolutionary trade-off between defence against grazing and competitive ability in a simple unicellular alga, *Chlorella vulgaris*. *Proc. R. Soc. B Biol. Sci.* 271:1947–53
- Zandona E, Auer SK, Kilham SS, Howard JL, López-Sepulcre A, et al. 2011. Diet quality and prey selectivity correlate with life histories and predation regime in Trinidadian guppies. *Funct. Ecol.* 25:964–73